

Zaraius Bilimoria

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EDUCATION

Olin College of Engineering

B.S. in Robotics Engineering, GPA: 3.97

Needham, MA

May 2028

EXPERIENCE

Robotics Intern

Jun 2026 – Aug 2026

HITT Contracting

Falls Church, VA

- Architected a ROS 2 social navigation system for Boston Dynamics Spot, integrating human perception, prediction, and Nav2 motion planning for operation around pedestrians on active construction sites
- Developed a C++ ROS 2 perception pipeline using ZED 2i skeleton tracking to estimate pedestrian position, velocity, and posture classification from 38 body keypoints, providing real-time human state information for navigation
- Implemented a custom Nav2 MPPI SocialCritic plugin in C++ that predicts pedestrian trajectories and applies anisotropic Gaussian cost fields across 5,000 candidate trajectories per planning cycle, enabling real-time socially-aware navigation around stationary and moving pedestrians
- Validated the full autonomy stack in a live on-site demo on an active construction site, integrating perception, localization, and navigation and resolving TF, sensor synchronization, and coordinate-frame issues on Boston Dynamics Spot
- Built a Docker-based TurtleBot3 simulation environment for rapid development and repeatable validation of the autonomy stack before deployment to the physical robot

Embedded Systems & Computer Vision Intern

Jun 2025 – Dec 2025

Prophet AI

Somerville, MA

- Designed and deployed a scalable 12-camera network for a 100-bird poultry farm using Raspberry Pis, replacing a legacy IP-camera system; programmed H.264-over-RTSP stream in Python, ensuring 24/7 uptime with automated IP provisioning
- Boosted inference throughput from 220 to 3,000 FPS and cut compute load by 75% by training a lightweight PyTorch YOLO model and building a custom pixel-based motion filter with OpenCV, allowing continuous poultry health monitoring
- Increased the model F1 score from 0.94 to 0.99 by implementing a dynamic masking system to improve data quality, and by developing an active-learning pipeline which trains two CV models nightly, computes IoU disagreement, and flags discrepancies for retraining on flagged images

Robotics Software Engineer

Aug 2024 – Current

Olin Agriculture Robotics Research Team

Needham, MA

- Leading an autonomous ground robot for soil moisture mapping, developing crop-row segmentation and a ROS 2/Nav2 navigation system for autonomous field navigation
- Designed and implemented ROS 2 perception and sensor fusion algorithms for real-time crop-row detection and global pose estimation, enabling autonomous navigation and crop-row alignment in unstructured environments
- Developed 3D Gaussian splatting techniques to process depth camera data, generating dense 3D point clouds for weed identification and path planning
- Designed and fabricated a protoboard-soldered signal-conditioning interface for a commercial resistive soil probe

Software & Integration Lead

Aug 2024 – May 2026

Olin AERO

Needham, MA

- Taught core robotics software concepts (ROS 2, state machines, computer vision, and networking) to 6 first-year team members, structuring mini-lectures and projects to ensure contributions to the SUAS competition aircraft
- Developed and led the implementation of the communication system for an autonomous aerial vehicle using TCP to reliably transmit live images and flight data from onboard Pixhawk autopilot and Jetson Nano to a ground station
- Integrated Pixhawk autopilot for bidirectional control and data flow with onboard computing systems

Captain, Lead Programmer – NUTRONS 125

Jan 2019 – Jul 2024

FIRST Robotics

Revere, MA

- Led a 19-member team to create a robot ranked 2nd out of 600+ teams at the global championship
- Developed autonomous and teleoperated robot code in Java, collaborating with mentors and the mechanical team

TECHNICAL SKILLS

Robotics & Embedded Systems: ROS 2, Nav2, Docker, Gazebo, RViz, Foxglove, Ubuntu, Raspberry Pi, STM32

Programming & Development: C++, Python, C, Java, MATLAB, Git, CMake, Bash, Tmux, Vim, Julia

Computer Vision & ML: PyTorch, OpenCV, YOLO, TensorRT, ZED SDK, Open3D, DeepStream SDK

Design & Fabrication: Onshape, SolidWorks, KiCad, 3D Printing, Soldering, Hand Tools, Power Tools